



Fallstudie – Pine Valley Furniture Company

Modern Database Management (Hoffer, 2016)

Ausschnitt aus Kapitel 1 - *Database Environment and Development Process*

DEVELOPING A DATABASE APPLICATION FOR PINE VALLEY FURNITURE COMPANY

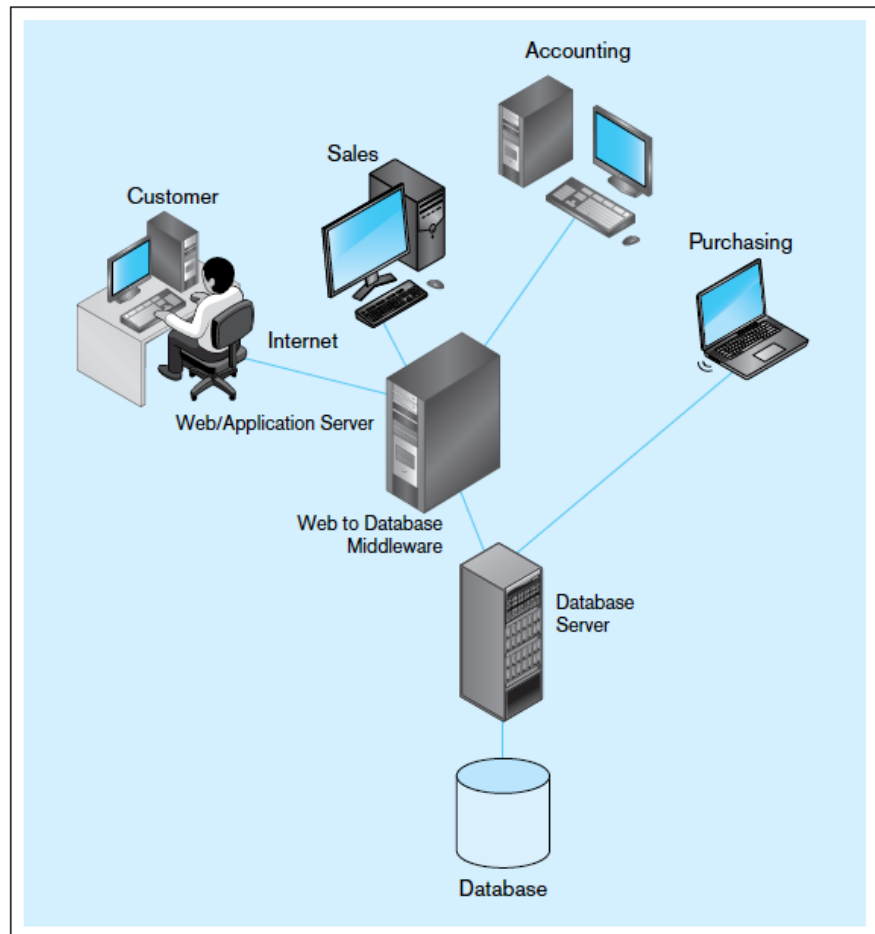


Pine Valley Furniture Company was introduced earlier in this chapter. By the late 1990s, competition in furniture manufacturing had intensified, and competitors seemed to respond more rapidly than Pine Valley Furniture to new business opportunities. While there were many reasons for this trend, managers believed that the computer information systems they had been using (based on traditional file processing) had become outdated. After attending an executive development session led by Heikki Topi and Jeff Hoffer (we wish!), the company started a development effort that eventually led to adopting a database approach for the company. Data previously stored in separate files have been integrated into a single database structure. Also, the metadata that describe these data reside in the same structure. The DBMS provides the interface between the various database applications for organizational users and the database (or databases). The DBMS allows users to share the data and to query, access, and update the stored data.

To facilitate the sharing of data and information, Pine Valley Furniture Company uses a local area network (LAN) that links employee workstations in the various departments to a database server, as shown in Figure 1-13. During the early 2000s, the company mounted a two-phase effort to introduce Internet technology. First, to improve intracompany communication and decision making, an intranet was installed that allows employees fast Web-based access to company information, including phone directories, furniture design specifications, e-mail, and so forth. In addition, Pine Valley Furniture Company also added a Web interface to some of its business applications, such as order entry, so that more internal business activities that require access to data in the database server could also be conducted by employees through



FIGURE 1-13 Computer System for Pine Valley Furniture Company



its intranet. However, most applications that use the database server still do not have a Web interface and require that the application itself be stored on employees' workstations.

Database Evolution at Pine Valley Furniture Company

A trait of a good database is that it does and can evolve! Helen Jarvis, product manager for home office furniture at Pine Valley Furniture Company, knows that competition has become fierce in this growing product line. Thus, it is increasingly important to Pine Valley Furniture that Helen be able to analyze sales of her products more thoroughly. Often these analyses are ad hoc, driven by rapidly changing and unanticipated business conditions, comments from furniture store managers, trade industry gossip, or personal experience. Helen has requested that she be given direct access to sales data with an easy-to-use interface so that she can search for answers to the various marketing questions she will generate.

Chris Martin is a systems analyst in Pine Valley Furniture's information systems development area. Chris has worked at Pine Valley Furniture for five years and has experience with information systems from several business areas within Pine Valley. With this experience, his information systems education at Western Florida University, and the extensive training Pine Valley has given him, he has become one of Pine Valley's best systems developers. Chris is skilled in data modeling and is familiar with several relational database management systems used within the firm. Because of his experience, expertise, and availability, the head of information



systems has assigned Chris to work with Helen on her request for a marketing support system.

Because Pine Valley Furniture has been careful in the development of its systems, especially since adopting the database approach, the company already has databases that support its operational business functions. Thus, it is likely that Chris will be able to extract the data Helen needs from existing databases. Pine Valley's information systems architecture calls for systems such as the one Helen is requesting to be built as stand-alone databases so that the unstructured and unpredictable use of data will not interfere with the access to the operational databases needed to support efficient transaction processing systems.

Further, because Helen's needs are for data analysis, not creation and maintenance, and are personal, not institutional, Chris decides to follow a combination of prototyping and life-cycle approaches in developing the system Helen has requested. This means that Chris will follow all the life-cycle steps but focus his energy on the steps that are integral to prototyping. Thus, he will quickly address project planning and then use an iterative cycle of analysis, design, and implementation to work closely with Helen to develop a working prototype of the system she needs. Because the system will be personal and likely will require a database with limited scope, Chris hopes the prototype will end up being the actual system Helen will use. Chris has chosen to develop the system using Microsoft Access, Pine Valley's preferred technology for personal databases.

Project Planning

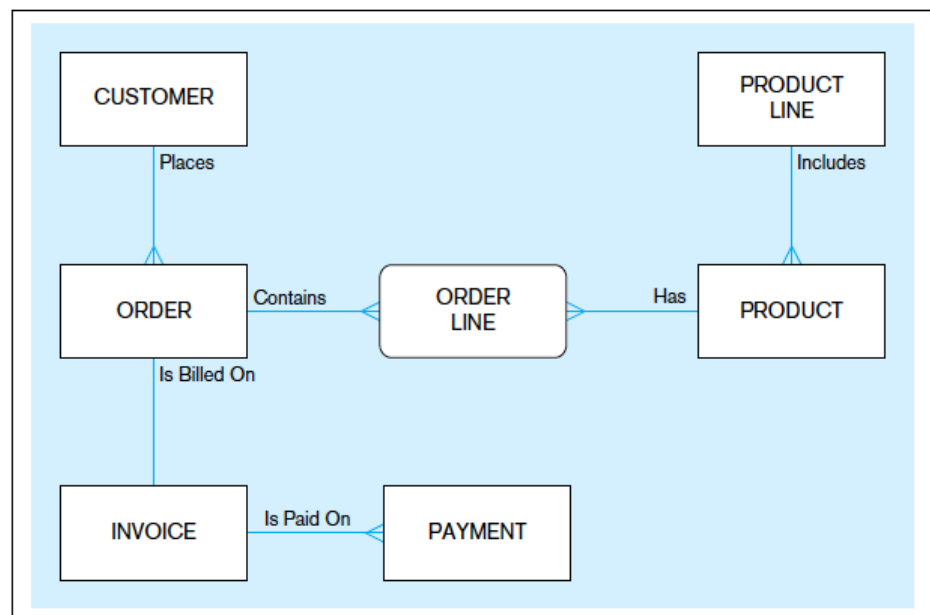
Chris begins the project by interviewing Helen. Chris asks Helen about her business area, taking notes about business area objectives, business functions, data entity types, and other business objects with which she deals. At this point, Chris listens more than he talks so that he can concentrate on understanding Helen's business area; he interjects questions and makes sure that Helen does not try to jump ahead to talk about what she thinks she needs with regards to computer screens and reports from the information system. Chris asks general questions, using business and marketing terminology as much as possible. For example, Chris asks Helen what issues she faces managing the home office products; what people, places, and things are of interest to her in her job; how far back in time she needs data to go to do her analyses; and what events occur in the business that are of interest to her. Chris pays particular attention to Helen's objectives as well as the data entities that she is interested in.

Chris does two quick analyses before talking with Helen again. First, he identifies all of the databases that contain data associated with the data entities Helen mentioned. From these databases, Chris makes a list of all of the data attributes from these data entities that he thinks might be of interest to Helen in her analyses of the home office furniture market. Chris's previous involvement in projects that developed Pine Valley's standard sales tracking and forecasting system and cost accounting system helps him speculate on the kinds of data Helen might want. For example, the objective to exceed sales goals for each product finish category of office furniture suggests that Helen wants product annual sales goals in her system; also, the objective of achieving at least an 8 percent annual sales growth means that the prior year's orders for each product need to be included. He also concludes that Helen's database must include all products, not just those in the office furniture line, because she wants to compare her line to others. However, he is able to eliminate many of the data attributes kept on each data entity. For example, Helen does not appear to need various customer data such as address, phone number, contact person, store size, and salesperson. Chris does, though, include a few additional attributes, customer type and zip code, which he believes might be important in a sales forecasting system.

Second, from this list, Chris draws a conceptual data model (Figure 1-14) that represents the data entities with the associated data attributes, as well as the major relationships among these data entities. The data model is represented using a notation called the Entity-Relationship (E-R) model. You will learn more about this notation in



FIGURE 1-14 Preliminary data model for Home Office product line marketing support system



Chapters 2 and 3. The data attributes of each entity Chris thinks Helen wants for the system are listed in Table 1-6. Chris lists in Table 1-6 only basic data attributes from existing databases, because Helen will likely want to combine these data in various ways for the analyses she will want to do.

Analyzing Database Requirements

Prior to their next meeting, Chris sends Helen a rough project schedule outlining the steps he plans to follow and the estimated length of time each step will take. Because prototyping is a user-driven process, in which the user says when to stop iterating on the new prototype versions, Chris can provide only rough estimates of the duration of certain project steps.

Chris does more of the talking at this second meeting, but he pays close attention to Helen's reactions to his initial ideas for the database application. He methodically walks through each data entity in Figure 1-14, explaining what it means and what business policies and procedures are represented by each line between entities.

A few of the rules he summarizes are listed here:

1. Each CUSTOMER *Places* any number of ORDERS. Conversely, each ORDER *Is Placed By* exactly one CUSTOMER.
2. Each ORDER *Contains* any number of ORDER LINES. Conversely, each ORDER LINE *Is Contained In* exactly one ORDER.
3. Each PRODUCT *Has* any number of ORDER LINES. Conversely, each ORDER LINE *Is For* exactly one PRODUCT.
4. Each ORDER *Is Billed On* one INVOICE and each INVOICE *Is a Bill for* exactly one ORDER.

Places, *Contains*, and *Has* are called one-to-many relationships because, for example, one customer places potentially many orders and one order is placed by exactly one customer.

In addition to the relationships, Chris also presents Helen with some detail on the data attributes captured in Table 1-6. For example, Order Number uniquely identifies each order. Other data about an order Chris thinks Helen might want to know include


TABLE 1-6 Data Attributes for Entities in the Preliminary Data Model (Pine Valley Furniture Company)

Entity Type	Attribute
Customer	Customer Identifier
	Customer Name
	Customer Type
	Customer Zip Code
Product	Product Identifier
	Product Description
	Product Finish
	Product Price
	Product Cost
	Product Annual Sales Goal
	Product Line Name
Product Line	Product Line Name
	Product Line Annual Sales Goal
Order	Order Number
	Order Placement Date
	Order Fulfillment Date
	Customer Identifier
Ordered Product	Order Number
	Product Identifier
	Order Quantity
Invoice	Invoice Number
	Order Number
	Invoice Date
Payment	Invoice Number
	Payment Date
	Payment Amount

the date when the order was placed and the date when the order was filled. (This would be the latest shipment date for the products on the order.) Chris also explains that the Payment Date attribute represents the most recent date when the customer made any payments, in full or partial, for the order.

Maybe because Chris was so well prepared or so enthusiastic, Helen is excited about the possibilities, and this excitement leads her to tell Chris about some additional data she wants (the number of years a customer has purchased products from Pine Valley Furniture Company and the number of shipments necessary to fill each order). Helen also notes that Chris has only one year of sales goals indicated for a product line. She reminds him that she wants these data for both the past and current years. As she reacts to the data model, Chris asks her how she intends to use the data she wants. Chris does not try to be thorough at this point because he knows that Helen has not worked with an information set like the one being developed; thus, she may not yet be positive about what data she wants or what she wants to do with it. Rather, Chris's objective is to understand a few ways in which Helen intends to use the data so he can develop an initial prototype, including the database and several computer displays or reports. The final list of attributes that Helen agrees she needs appears in Table 1-7.


**TABLE 1-7 Data Attributes for Entities in Final Data Model
(Pine Valley Furniture Company)**

Entity Type	Attribute
Customer	Customer Identifier
	Customer Name
	Customer Type
	Customer Zip Code
	<i>Customer Years</i>
Product	Product Identifier
	Product Description
	Product Finish
	Product Price
	Product Cost
Product Line	<i>Product Prior Year Sales Goal</i>
	<i>Product Current Year Sales Goal</i>
	Product Line Name
	Product Line Name
	<i>Product Line Prior Year Sales Goal</i>
Order	<i>Product Line Current Year Sales Goal</i>
	Order Number
	Order Placement Date
	Order Fulfillment Date
	<i>Order Number of Shipments</i>
Ordered Product	Customer Identifier
	Order Number
	Product Identifier
Invoice	Order Quantity
	Invoice Number
	Order Number
Payment	Invoice Date
	Invoice Number
	Payment Date
	Payment Amount

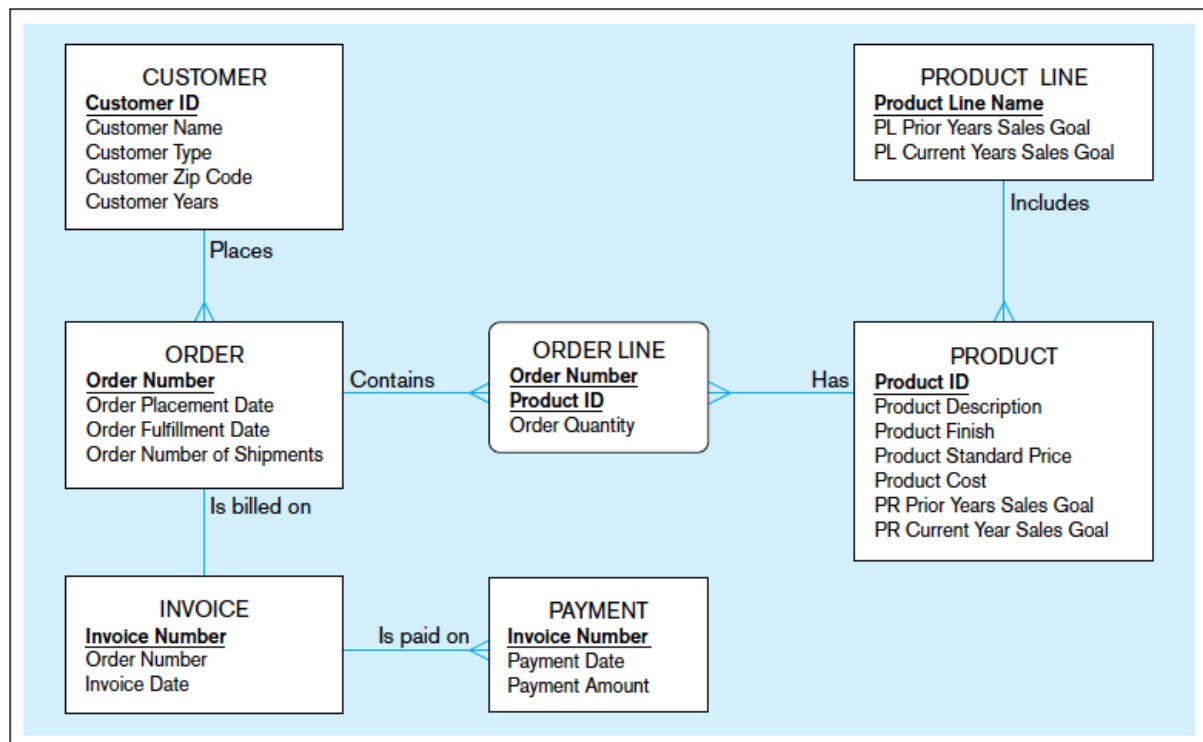
*Changes from preliminary list of attributes appear in italics.

Designing the Database

Because Chris is following a prototyping methodology and the first two sessions with Helen quickly identified the data Helen might need, Chris is now ready to build a prototype. His first step is to create a project data model like the one shown in Figure 1-15. Notice the following characteristics of the project data model:

1. It is a model of the organization that provides valuable information about how the organization functions, as well as important constraints.
2. The project data model focuses on entities, relationships, and business rules. It also includes attribute labels for each piece of data that will be stored in each entity.

Second, Chris translates the data model into a set of tables for which the columns are data attributes and the rows are different sets of values for those attributes.

**FIGURE 1-15** Project data model for Home Office product line marketing support system

Tables are the basic building blocks of a relational database (we will learn about this in Chapter 4), which is the database style for Microsoft Access. Figure 1-16 shows four tables with sample data: Customer, Product, Order, and OrderLine. Notice that these tables represent the four entities shown in the project data model (Figure 1-15). Each column of a table represents an attribute (or characteristic) of an entity. For example, the attributes shown for Customer are CustomerID and CustomerName. Each row of a table represents an instance (or occurrence) of the entity. The design of the database also required Chris to specify the format, or properties, for each attribute (MS Access calls attributes *fields*). These design decisions were easy in this case because most of the attributes were already specified in the corporate data dictionary.

The tables shown in Figure 1-14 were created using SQL (you will learn about this in Chapters 6 and 7). Figures 1-17 and 1-18 show the SQL statements that Chris would have likely used to create the structure of the ProductLine and Product tables. It is customary to add the suffix *_T* to a table name. Also note that because Access does not allow for spaces between names, the individual words in the attributes from the data model have now been concatenated. Hence, Product Description in the data model has become ProductDescription in the table. Chris did this translation so that each table had an attribute, called the table's "primary key," which will be distinct for each row in the table. The other major properties of each table are that there is only one value for each attribute in each row; if we know the value of the identifier, there can be only one value for each of the other attributes. For example, for any product line, there can be only one value for the current year's sales goal.

A final key characteristic of the relational model is that it represents relationships between entities by values stored in the columns of the corresponding tables. For example, notice that CustomerID is an attribute of both the Customer table and the Order table. As a result, we can easily link an order to its associated customer. For example, we can determine that OrderID 1003 is associated with CustomerID 1. Can you determine which ProductIDs are associated with OrderID 1004?

**FIGURE 1-16** Four relations (Pine Valley Furniture Company)

(a) Order and Order Line Tables

OrderID	OrderDate	CustomerID
1001	10/21/2015	1
1002	10/21/2015	8
1003	10/22/2015	15
1004	10/22/2015	5
1005	10/24/2015	3
1006	10/24/2015	2
1007	10/27/2015	11
1008	10/30/2015	12
1009	11/5/2015	4
1010	11/5/2015	1

OrderID	ProductID	OrderedQuantity
1001	1	2
1001	2	2
1001	4	1
1002	3	5
1003	3	3
1004	6	2
1004	8	2
1005	4	4
1006	4	1
1006	5	2
1006	7	2
1007	1	3
1007	2	2
1008	3	3
1008	8	3
1009	4	2
1009	7	3
1010	8	10

CustomerID	CustomerName
1	Contemporary Casuals
2	Value Furniture
3	Home Furnishings
4	Eastern Furniture
5	Impressions
6	Furniture Gallery
7	Period Furniture
8	California Classics
9	M and H Casual Furniture
10	Seminole Interiors
11	American Euro Lifestyles
12	Battle Creek Furniture
13	Heritage Furnishings
14	Kaneohe Homes
15	Mountain Scenes

ProductID	ProductDescription	ProductMaterial
1	End Table	Cherry
2	Coffee Table	Natural
3	Computer Desk	Natural
4	Entertainment Center	Natural
5	Writers Desk	Cherry
6	8-Drawer Desk	White
7	Dining Table	Natural
8	Computer Desk	Walnut

(b) Customer table

(c) Product table

In Chapters 6 and 7, you will also learn how to retrieve data from these tables by using SQL, which exploits these linkages. The other major decision Chris has to make about database design is how to physically organize the database to respond as fast as possible to the queries Helen will write. Because the database will be used for decision support, neither Chris nor Helen can anticipate all of the queries that will arise; thus, Chris must make the physical design choices from experience rather than precise knowledge of the way the database will be used. The key physical



```
CREATE TABLE ProductLine_T
(ProductLineID    VARCHAR (40) NOT NULL PRIMARY KEY,
PIPriorYearGoal  DECIMAL,
PICurrentYearGoal DECIMAL);
```

FIGURE 1-17 SQL definition of ProductLine table

```
CREATE TABLE Product_T
(ProductID        NUMBER(11,0) NOT NULL PRIMARY KEY
ProductDescription VARCHAR (50),
ProductFinish     VARCHAR (20),
ProductStandardPrice DECIMAL(6,2),
ProductCost       DECIMAL,
ProductPriorYearGoal DECIMAL,
ProductCurrentYearGoal DECIMAL,
ProductLineID     VARCHAR (40),
FOREIGN KEY       (ProductLineID) REFERENCES ProductLine_T (ProductLineID));
```

FIGURE 1-18 SQL definition of Product table

database design decision that SQL allows a database designer to make is on which attributes to create indexes. All primary key attributes (such as OrderNumber for the Order_T table)—those with unique values across the rows of the table—are indexed. In addition to this, Chris uses a general rule of thumb: Create an index for any attribute that has more than 10 different values and that Helen might use to segment the database. For example, Helen indicated that one of the ways she wants to use the database is to look at sales by product finish. Thus, it might make sense to create an index on the Product_T table using the Product Finish attribute.

However, Pine Valley uses only six product finishes, or types of wood, so this is not a useful index candidate. On the other hand, OrderPlacementDate (called a secondary key because there may be more than one row in the Order_T table with the same value of this attribute), which Helen also wants to use to analyze sales in different time periods, is a good index candidate.

Using the Database

Helen will use the database Chris has built mainly for ad hoc questions, so Chris will train her so that she can access the database and build queries to answer her ad hoc questions. Helen has indicated a few standard questions she expects to ask periodically. Chris will develop several types of prewritten routines (forms, reports, and queries) that can make it easier for Helen to answer these standard questions (so she does not have to program these questions from scratch).

During the prototyping development process, Chris may develop many examples of each of these routines as Helen communicates more clearly what she wants the system to be able to do. At this early stage of development, however, Chris wants to develop one routine to create the first prototype. One of the standard sets of information Helen says she wants is a list of each of the products in the Home Office product line showing each product's total sales to date compared with its current year sales goal. Helen may want



FIGURE 1-19 SQL query for Home Office sales-to-goal comparison

```

SELECT Product.ProductID, Product.ProductDescription, Product.PRCurrentYearSalesGoal,
       (OrderQuantity * ProductPrice) AS SalesToDate
FROM Order.OrderLine, Product.ProductLine
WHERE Order.OrderNumber = OrderLine.OrderNumber
AND Product.ProductID = OrderedProduct.ProductID
AND Product.ProductID = ProductLine.ProductID
AND Product.ProductLineName = "Home Office";

```

FIGURE 1-20 Home Office product line sales comparison

Home Office Sales to Date : Select Query				
	Product ID	Product Description	PR Current Year Sales Goal	Sales to Date
	3	Computer Desk	\$23,500.00	5625
	10	96" Bookcase	\$22,500.00	4400
	5	Writer's Desk	\$26,500.00	650
	3	Computer Desk	\$23,500.00	3750
	7	48" Bookcase	\$17,000.00	2250
	5	Writer's Desk	\$26,500.00	3900

the results of this query to be displayed in a more stylized fashion—an opportunity to use a report—but for now Chris will present this feature to Helen only as a query.

The query to produce this list of products appears in Figure 1-19, with sample output in Figure 1-20. The query in Figure 1-19 uses SQL. You can see three of the six standard SQL clauses in this query: SELECT, FROM, and WHERE. SELECT indicates which attributes will be shown in the result. One calculation is also included and given the label “Sales to Date.” FROM indicates which tables must be accessed to retrieve data. WHERE defines the links between the tables and indicates that results from only the Home Office product line are to be included. Only limited data are included for this example, so the Total Sales results in Figure 1-20 are fairly small, but the format is the result of the query in Figure 1-19.

Chris is now ready to meet with Helen again to see if the prototype is beginning to meet her needs. Chris shows Helen the system. As Helen makes suggestions, Chris is able to make a few changes online, but many of Helen’s observations will have to wait for more careful work at his desk.

Space does not permit us to review the whole project to develop the Home Office marketing support system. Chris and Helen ended up meeting about a dozen times before Helen was satisfied that all the attributes she needed were in the database; that the standard queries, forms, and reports Chris wrote were of use to her; and that she knew how to write queries for unanticipated questions. Chris will be available to Helen at any time to provide consulting support when she has trouble with the system, including writing more complex queries, forms, or reports. One final decision that Chris and Helen made was that the performance of the final prototype was efficient enough that the prototype did not have to be rewritten or redesigned. Helen was now ready to use the system.

Administering the Database

The administration of the Home Office marketing support system is fairly simple. Helen decided that she could live with weekly downloads of new data from Pine Valley’s



operational databases into her MS Access database. Chris wrote a C# program with SQL commands embedded in it to perform the necessary extracts from the corporate databases and wrote an MS Access program in Visual Basic to rebuild the Access tables from these extracts; he scheduled these jobs to run every Sunday evening. Chris also updated the corporate information systems architecture model to include the Home Office marketing support system. This step was important so that when changes occurred to formats for data included in Helen's system, the corporate data modeling and design tools could alert Chris that changes might also have to be made in her system.

Future of Databases at Pine Valley

Although the databases currently in existence at Pine Valley adequately support the daily operations of the company, requests such as the one made by Helen have highlighted that the current databases are often inadequate for decision support applications. For example, following are some types of questions that cannot be easily answered:

1. What is the pattern of furniture sales this year, compared with the same period last year?
2. Who are our 10 largest customers, and what are their buying patterns?
3. Why can't we easily obtain a consolidated view of any customer who orders through different sales channels, rather than viewing each contact as representing a separate customer?

To answer these and other questions, an organization often needs to build a separate database that contains historical and summarized information. Such a database is usually called a *data warehouse* or, in some cases, a *data mart*. Also, analysts need specialized decision support tools to query and analyze the database. One class of tools used for this purpose is called online analytical processing (OLAP) tools. We describe data warehouses, data marts, and related decision support tools in Chapter 11. There you will learn of the interest in building a data warehouse that is now growing within Pine Valley Furniture Company.